Week 6 Double Machine Learning for Partially Linear IV Model

Partially Linear IV Model

We consider the partially linear structural equation model

$$\begin{aligned} Y_i &:= \theta_0 D_i + g_0(X_i) + \zeta_i, & E[\zeta_i \mid Z_i, X_i] = 0 \\ Z_i &:= m_0(X_i) + \epsilon_i, & E[\epsilon_i \mid X_i] = 0 \end{aligned}$$

where there is a structural or causal relation between D_i and Y_i, captured by θ_0. $g_0(X_i)+\zeta_i$ is the stochastic error, partly explained by covariates X_i. ϵ_i and ζ_i are stochastic errors that are not explained by X_i. Y_i and D_i are jointly determined, meaning that D_i is *endogeneous*, with $E[\zeta_i \mid D_i] \neq 0$. The instrument, Z_i, is introduced to create exogenous variation in D_i. The observed X_i serve as confounding factors, and we can think of variation in Z_i as being exogenous conditional on X_i.

PLIV Model in Residualized Form

The PLIV model above can be rewritten in the following residualized form

$$\tilde{Y}_i = \theta_0 \tilde{D}_i + \zeta_i, \quad E[\zeta_i \mid \epsilon_i, X_i] = 0$$

where

$$\begin{aligned} \tilde{Y}_i &= (Y_i - \ell_0(X_i)), & \ell_0(X_i) = E[Y_i \mid X_i] \\ \tilde{D}_i &= (D_i - r_0(X_i)), & r_0(X_i) = E[D_i \mid X_i] \\ \tilde{Z}_i &= (Z_i - m_0(X_i)), & m_0(X_i) = E[Z_i \mid X_i] \end{aligned}$$

DML for PLIV Model

Given the valid instrument for identification, DML proceeds as follows.

Compute the estimates $\hat{\ell}_0$, $\hat{r}_0$, and $\hat{m}_0$, which amounts to solving the three problems of predicting Y_i, D_i, and Z_i using X_i, using any generic ML methods. Then estimate θ_0 by the the instrumental variable regression of $\tilde{Y}$ on $\tilde{D}$ using $\tilde{Z}$ as an instrument. Use the conventional inference for the IV regression estimator, ignoring the estimation error in these residuals.

```r
library(hdm)
library(AER)
library(randomForest)
library(lfe)
library(glmnet)
set.seed(1)
```

```r
## DML for PLIV Model
dml2_for_plivm <- function(x, d, z, y, dreg, yreg, zreg, nfold = 5) {

  ## cross-fitting
  nobs <- nrow(x)
  foldid <- rep.int(1:nfold, times = ceiling(nobs / nfold))[sample.int(nobs)]
  I <- split(1:nobs, foldid)

  ## create residualized objects to fill
  ytil <- dtil <- ztil <- rep(NA, nobs)

  ## obtain cross-fitted residuals
  cat("fold: ")
  for (b in seq_along(I)) {
    dfit <- dreg(x[-I[[b]], ], d[-I[[b]]]) ## take a fold out
    zfit <- zreg(x[-I[[b]], ], z[-I[[b]]]) ## take a fold out
    yfit <- yreg(x[-I[[b]], ], y[-I[[b]]]) ## take a fold out
    dhat <- predict(dfit, x[I[[b]], ], type = "response") ## predict the held-out fold
    zhat <- predict(zfit, x[I[[b]], ], type = "response") ## predict the held-out fold
    yhat <- predict(yfit, x[I[[b]], ], type = "response") ## predict the held-out fold
    dtil[I[[b]]] <- (d[I[[b]]] - dhat) # record residual
    ztil[I[[b]]] <- (z[I[[b]]] - zhat) # record residual
    ytil[I[[b]]] <- (y[I[[b]]] - yhat) # record residual
    cat(b, " ")
  }
  ivfit <- tsls(y = ytil, d = dtil, x = NULL, z = ztil, intercept = FALSE)
  coef_est <- ivfit$coef ## extract coefficient
  se <- ivfit$se ## record standard error
  cat(sprintf("\ncoef (se) = %g (%g)\n", coef_est, se))
```

```r
  return(list(coef_est = coef_est, se = se, dtil = dtil, ytil = ytil, ztil = ztil))
}
```

Emprical Example: Acemoglu, Johnson, Robinson (AER)

- Y_i is log modern GDP;
- D_i is a measure of protection from expropriation, a proxy for quality of institutions;
- Z_i is the log of settler's mortality;
- W_i are geographical variables (latitude, latitude squared, continent dummies as well as interactions)

```r
## read AJR data
file <- "https://raw.githubusercontent.com/CausalAIBook/MetricsMLNotebooks/main/data/AJ
AJR <- read.csv(file)
dim(AJR)
```

```
[1] 64 11
```

```r
## construct variables
y <- AJR$GDP
d <- AJR$Exprop
z <- AJR$logMort
xraw <- model.matrix(~ Latitude + Africa + Asia + Namer + Samer, data = AJR)
x <- model.matrix(~ -1 + (Latitude + Latitude2 + Africa + Asia + Namer + Samer)^2, data
dim(x)
```

```
[1] 64 21
```

```r
## IV specification and weak IV test
library(ivreg)
fit <- ivreg(y ~ d + Latitude + Africa + Asia + Namer + Samer |
               z + Latitude + Africa + Asia + Namer + Samer,
             data = AJR)
summary(fit, diagnostics = TRUE)
```

```
Call:
```

```
ivreg(formula = y ~ d + Latitude + Africa + Asia + Namer + Samer |
    z + Latitude + Africa + Asia + Namer + Samer, data = AJR)

Residuals:
     Min       1Q   Median       3Q      Max
-2.56827 -0.59648 -0.06903  0.79095  1.91474

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.0084     3.2492   0.310   0.7574
d             1.0360     0.4100   2.527   0.0143 *
Latitude     -1.0099     1.4325  -0.705   0.4837
Africa        0.4112     1.1476   0.358   0.7215
Asia         -0.1300     0.8071  -0.161   0.8726
Namer         0.9252     0.9658   0.958   0.3421
Samer         0.8040     0.9095   0.884   0.3804

Diagnostic tests:
                 df1 df2 statistic p-value
Weak instruments   1  57     3.846 0.05476 .
Wu-Hausman         1  56     8.432 0.00527 **
Sargan             0  NA        NA      NA
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.012 on 57 degrees of freedom
Multiple R-Squared: 0.1499, Adjusted R-squared: 0.06037
Wald test: 6.448 on 6 and 57 DF,  p-value: 3.147e-05
```

```r
set.seed(1)

## DML with Post-Lasso
cat(sprintf("\n DML with Post-Lasso \n"))
```

```
 DML with Post-Lasso
```

```r
dreg <- function(x, d) {
  hdm::rlasso(x, d, post=TRUE) ## default plug-in penalty level
}
yreg <- function(x, y) {
  hdm::rlasso(x, y, post=TRUE) ## default plug-in penalty level
}
zreg <- function(x, z) {
  hdm::rlasso(x, z, post=TRUE) ## default plug-in penalty level
}

## specify the function
dml2_lasso <- dml2_for_plivm(x, d, z, y, dreg, yreg, zreg, nfold = 5)
```

```
fold: 1  2  3  4  5
coef (se) = 0.731405 (0.185907)
```

```r
# DML with Random Forest
cat(sprintf("\n DML with Random Forest \n"))
```

```
 DML with Random Forest
```

```r
dreg <- function(x, d) {
  randomForest(x, d) ## default ntrees = 500
}
yreg <- function(x, y) {
  randomForest(x, y) ## default ntrees = 500
}
zreg <- function(x, z) {
  randomForest(x, z) ## default ntrees = 500
}

## specify the function
dml2_rf <- dml2_for_plivm(xraw, d, z, y, dreg, yreg, zreg, nfold = 5)
```

```
fold: 1  2  3  4  5
coef (se) = 0.857695 (0.332819)
```

```r
## compare Forest vs Lasso
comp_tab <- matrix(NA, 3, 2)
comp_tab[1, ] <- c(sqrt(mean((dml2_rf$ytil)^2)), sqrt(mean((dml2_lasso$ytil)^2)))
comp_tab[2, ] <- c(sqrt(mean((dml2_rf$dtil)^2)), sqrt(mean((dml2_lasso$dtil)^2)))
comp_tab[3, ] <- c(sqrt(mean((dml2_rf$ztil)^2)), sqrt(mean((dml2_lasso$ztil)^2)))
rownames(comp_tab) <- c("RMSE for Y:", "RMSE for D:", "RMSE for Z:")
colnames(comp_tab) <- c("RF", "LASSO")
print(comp_tab, digits = 3)
```

```
               RF LASSO
RMSE for Y: 0.766 0.828
RMSE for D: 1.259 1.554
RMSE for Z: 0.929 1.101
```

Weak Instrument Problem

```r
## using lfe package
summary(felm(dml2_lasso$dtil ~ dml2_lasso$ztil), robust = TRUE)
```

```

Call:
   felm(formula = dml2_lasso$dtil ~ dml2_lasso$ztil)

Residuals:
    Min      1Q  Median      3Q     Max
-3.2882 -1.0124  0.0809  1.0780  2.9289

Coefficients:
                Estimate Robust s.e t value Pr(>|t|)
(Intercept)     -0.04663    0.18288  -0.255   0.7996
dml2_lasso$ztil -0.52826    0.20859  -2.532   0.0139 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.464 on 62 degrees of freedom
Multiple R-squared(full model): 0.1402   Adjusted R-squared: 0.1263
```

```
Multiple R-squared(proj model): 0.1402   Adjusted R-squared: 0.1263
F-statistic(full model, *iid*):10.11 on 1 and 62 DF, p-value: 0.002304
F-statistic(proj model): 6.414 on 1 and 62 DF, p-value: 0.01387
```

```
summary(felm(dml2_rf$dtil ~ dml2_rf$ztil), robust = TRUE)
```

```

Call:
   felm(formula = dml2_rf$dtil ~ dml2_rf$ztil)

Residuals:
    Min      1Q  Median      3Q     Max
-2.4877 -1.0628  0.0102  0.7681  3.2911

Coefficients:
             Estimate Robust s.e t value Pr(>|t|)
(Intercept)   0.01889    0.15485   0.122   0.9033
dml2_rf$ztil -0.33538    0.18283  -1.834   0.0714 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.239 on 62 degrees of freedom
Multiple R-squared(full model): 0.06126   Adjusted R-squared: 0.04612
Multiple R-squared(proj model): 0.06126   Adjusted R-squared: 0.04612
F-statistic(full model, *iid*):4.046 on 1 and 62 DF, p-value: 0.04862
F-statistic(proj model): 3.365 on 1 and 62 DF, p-value: 0.0714
```

Anderson-Rubin Idea for Inference with Weak Instruments

As shown above, we may have weak instruments because the F-statistics in the regression $\tilde{D}_i \sim \tilde{Z}_i$ is small relative to standard rules of thumb – for example Stock and Yogo (2005) suggest accounting for weak instruments if the first stage F-statistic is less than 10 (and more recent work suggests even larger cutoffs).

Here, we consider one specific approach (from Anderson and Rubin (1949)) for valid inference under weak identification based upon the statistic:

$$C(\theta) = \frac{|\mathbb{E}_n[(\tilde{Y} - \theta\tilde{D})\tilde{Z}]|^2}{\mathbb{V}_n[(\tilde{Y} - \theta\tilde{D})\tilde{Z}]/n}$$

The empirical variance $\mathbb{V}_n$ is defined as

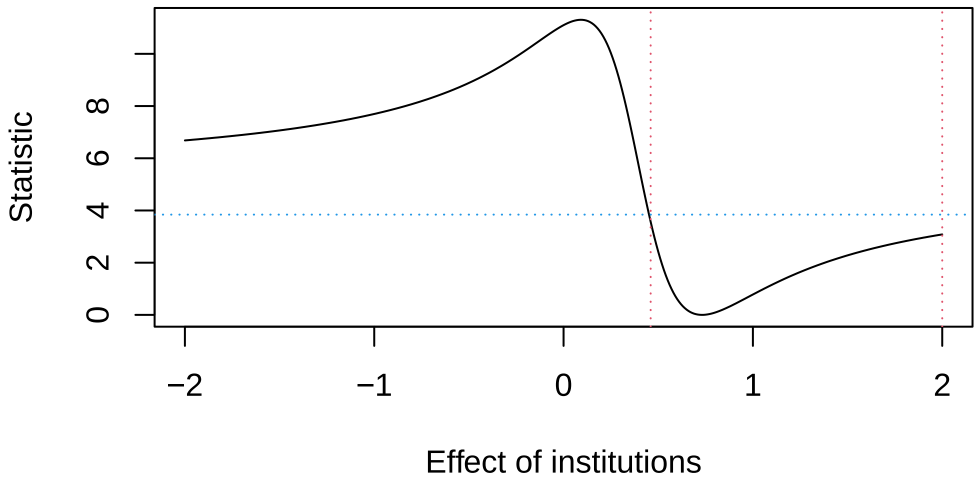
$$\mathbb{V}_n[g(W_i)] := \mathbb{E}_n g(W_i)g(W_i)' - \mathbb{E}_n[g(W_i)]\mathbb{E}_n[g(W_i)]'$$

If $\theta_0 = \theta$, then $C(\theta) \xrightarrow{d} \mathcal{N}(0,1)^2 = \chi^2(1)$. Therefore, we can reject the hypothesis $\theta_0 = \theta$ at level a (for example $a = .05$ for a 5% level test) if $C(\theta) > c(1-a)$ where $c(1-a)$ is the $(1-a)$- quantile of a $\chi^2(1)$ variable. The probability of falsely rejecting the true hypothesis is approximately $a \times 100\%$.

To construct a $(1-a) \times 100\%$ confidence region for θ, we can then simply invert the test by collecting all parameter values that are not rejected at the a level:

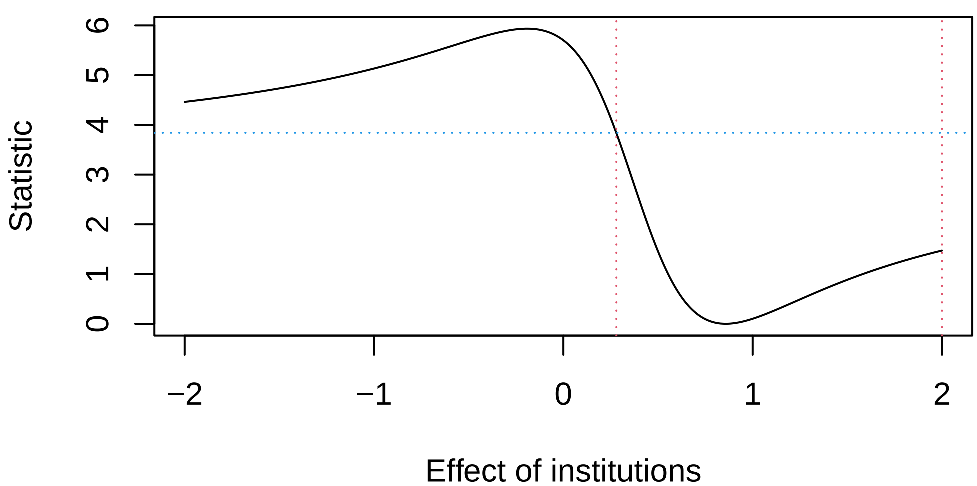
$$CR(\theta) = \{\theta \in \Theta : C(\theta) \leq c(1-a)\}$$

```r
## DML-AR (DML with Anderson-Rubin)
dml_ar_pliv <- function(rY, rD, rZ, grid, alpha = .05) {
  n <- length(rY)
  Cstat <- rep(0, length(grid))
  for (i in seq_along(grid)) {
    Cstat[i] <- n * (mean((rY - grid[i] * rD) * rZ))^2 / var((rY - grid[i] * rD) * rZ)
  }
  LB <- min(grid[Cstat < qchisq(1 - alpha, 1)])
  UB <- max(grid[Cstat < qchisq(1 - alpha, 1)])
  plot(range(grid), range(c(Cstat)), type = "n", xlab = "Effect of institutions", ylab
  lines(grid, Cstat, lty = 1, col = 1)
  abline(h = qchisq(1 - alpha, 1), lty = 3, col = 4)
  abline(v = LB, lty = 3, col = 2)
  abline(v = UB, lty = 3, col = 2)
  return(list(UB = UB, LB = LB))
}

dml_ar_pliv(
  rY = dml2_lasso$ytil, rD = dml2_lasso$dtil, rZ = dml2_lasso$ztil,
  grid = seq(-2, 2, by = .01)
)
```




```
$UB
[1] 2


$LB
[1] 0.46
```

```
dml_ar_pliv(
  rY = dml2_rf$ytil, rD = dml2_rf$dtil, rZ = dml2_rf$ztil,
  grid = seq(-2, 2, by = .01)
)
```




```
$UB
[1] 2
```

```
$LB
[1] 0.28
```

For any θ with $C(\theta) \leq 3.84$, we cannot reject the null hypothesis that $\theta = \theta_0$ at the 5% level. Those θ values form the Anderson–Rubin confidence set.